Q1. 1.

```
. regress wage yeduc nin
```

Source	SS	df	MS	Number of obs	=	61,136
				F(2, 61153)	=	4310.10
Model	3.2226e+14	2	1.6113e+14	Prob > F	=	0.0000
Residual	2.2812e+15	61,133	3.7315e+10	R-squared	=	0.1238
				Adj R-squared	=	0.1238
Total	2.6034e+15	61,135	4.2585e+10	Root MSE	=	1.9e+05

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	17826.42	193.9603	91.91	0.000	17446.26	18206.59
nin	-6880.068	714.7795	-9.63	0.000	-8281.038	-5479.099
_cons	72117.21	2519.881	28.62	0.000	67178.24	77056.19

$\hat{\beta}_1 = 17826.42$, Standard error of $\hat{\alpha}$ is 193.9603 , ∴ on average, an additional year of education increases the monthly wage by 17826.42

$\hat{\beta}_2 = -6880.068$, Standard error of $\hat{\beta}_1$ is 714.7795 ∴ on average, an additional number of inpres school per children decreases the monthly wage by 6880.068.

$\hat{\alpha} = 72117.21$, Standard error of $\hat{\beta}_2$ is 2519.881

∴ p-value of $\hat{\alpha}$ = 0.000 ∴ The regression estimates are statistically significant.

p-value of $\hat{\beta}_1$ = 0.000 The two variables have an important impact on the model.

p-value of $\hat{\beta}_2$ = 0.000

$R^2 = 0.1238$, it means that 12.38% of the variability in monthly wage can be explained by years of education and province-level number of inpres school per children

2.

```
. regress nin yeduc
```

Source	SS	df	MS	Number of obs	=	152,989
				F(1, 152987)	=	592.60
Model	762.477939	1	762.477939	Prob > F	=	0.0000
Residual	196043.137	152,987	1.28666578	R-squared	=	0.0039
				Adj R-squared	=	0.0039
Total	196805.615	152,988	1.29164127	Root MSE	=	1.1343

nin	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	-.017078	.0007015	-24.34	0.000	-.018453	-.015703
_cons	2.24942	.0063333	355.17	0.000	2.237006	2.261833

If we use the model $nin = \delta_0 + \delta_1 \cdot education + v$

According to the data, $\hat{\delta}_0 = 2.24942$ $\hat{\delta}_1 = -0.17078$

∵ p-value of $\hat{\delta}_0$ is 0.000 and p-value of $\hat{\delta}_1$ is 0.00

∴ The regression estimates is statistically significant.

nin and education are positively correlated.

∴ $wage_{ij} = \alpha + \beta_1 \cdot education_i + \beta_2 \cdot nin_j + \mu_i$

$\Rightarrow wage = \alpha + \beta_1 \cdot education + \beta_2 (\delta_0 + \delta_1 \cdot education + v) + \mu$

$= (\alpha + \beta_2 \delta_0) + (\beta_1 + \delta_1 \beta_2) education + (\mu + \beta_2 v)$

∴ Bias $(\tilde{\beta}_1) = \beta_2 \tilde{\delta}_1 = (-0.017078) \times (-6880.68) = 117.497201$

∵ $\hat{\delta}_0 > 0$, $\hat{\delta}_1 < 0$, $\hat{\beta}_2 < 0$

∴ For the $\tilde{\beta}_1$, there would be a positive bias. $\tilde{\beta}_1$ will be upward biased

From theoretical judgement

3. ∵ $\hat{\beta}_2 < 0$, Bias $(\tilde{\beta}_1) > 0$

∴ Cov(education$_i$, nin$_j$) < 0, direction of cov(education$_i$, nin$_j$) should be negative.

```
. correlate yeduc nin
(obs=152,989)
```

	yeduc	nin
yeduc	1.0000	
nin	-0.0621	1.0000

Based on data itself, the direction of cov(education$_i$, nin$_j$) is also negative

∴ The two results are consistent with each other.

```
. regress wage yeduc
```

Source	SS	df	MS	Number of obs	=	61,136
				F(1, 61134)	=	8530.77
Model	3.1880e+14	1	3.1880e+14	Prob > F	=	0.0000
Residual	2.2846e+15	61,134	3.7371e+10	R-squared	=	0.1225
				Adj R-squared	=	0.1224
Total	2.6034e+15	61,135	4.2585e+10	Root MSE	=	1.9e+05

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	17910.03	193.9109	92.36	0.000	17529.96	18290.09
_cons	57405.06	2004.097	28.63	0.000	53475.45	61334.66

$\tilde{\alpha} = 57405.06$

$\tilde{\beta}_1 = 17910.03$

∴ There would be omitting variable bias

For the $\tilde{\beta}_1$, there is a positive bias. $\tilde{\beta}_1$ is upward biased

Because wage is negatively correlated, when yeduc increases by unit, besides the impact of increasing yeduc on wage, nin also decreases with the increase of x, which also brings an increase in wage.

∴ The positive impact of nin on wage has been wrongly attributed to yeduc.

∴ $\tilde{\beta}_1$ is positively biased.

4. Model 3: $wage_{ij} = \alpha + \beta_1 \cdot education_i + \beta_2 nin_j + \beta_3 en71 + \mu_i$

```
. regress wage yeduc nin en71
```

Source	SS	df	MS	Number of obs	=	60,936
				F(3, 60932)	=	2873.13
Model	3.2216e+14	3	1.0739e+14	Prob > F	=	0.0000
Residual	2.2774e+15	60,932	3.7376e+10	R-squared	=	0.1239
				Adj R-squared	=	0.1239
Total	2.5996e+15	60,935	4.2661e+10	Root MSE	=	1.9e+05

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	17704.64	197.781	89.52	0.000	17316.98	18092.29
nin	-7513.557	735.1522	-10.22	0.000	-8954.457	-6072.656
en71	35557.16	9684.733	3.67	0.000	16575.06	54539.27
_cons	68233.96	2736.359	24.94	0.000	62870.69	73597.23

5. The corresponding hypothesis testing:

① The null hypothesis H_0: $\beta_3 = 0$

② The alternative H_1: $\beta_3 \neq 0$

③ We choose 5% as the significance value

④ According to the chart, we know that $t_{\beta_1} \approx 3.67$

⑤ Degree of freedom is n-k-1 = 152990 - 3 - 1 = 152986

$C_{0.05} = t_{0.05,\ 152986} \approx 1.96$

∴ $t_{\hat{\beta}_1} > C_{0.05}$

∴ The null hypothesis is rejected because the t-statistic exceeds the critical value

∴ We could take the alternative hypothesis H_1: $\beta_3 \neq 0$

∴ The enrollment rate in 1971 of each province does change the wage outcome.

6. F-test

```
. test nin en71

 ( 1)  nin = 0
 ( 2)  en71 = 0

       F(  2, 60932) =   53.17
            Prob > F =    0.0000
```

The Null Hypothesis H_0 : $\beta_2 = \beta_3 = 0$, coefficients of nin and en71 are all zero

The Alternative Hypothesis H_1 : H_0 is not true.

We choose the significance level as 1%

Through model $wage = \beta_0 + \beta_1 wage + \mu$ → we can get SSR_r

Through $wage = \beta_0 + \beta_1 wage + \beta_2 nin + \beta_3 en71 + \mu$ → we can get SSR_{ur}

$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n-k-1)} = 96.3246$$

If we choose significance level as 5%

∴ For $F_{2,60932}$, the critical value c = 53.17

∵ F > 53.17 ∴ we can reject H_0

∴ nin and en71 jointly affect wage.

Q2. 1. 1) linear probability model

```
. regress high_wage yeduc nin

      Source |       SS           df       MS      Number of obs   =    61,136
-------------+----------------------------------   F(2, 61133)     =   5721.30
       Model |  2409.73024         2  1204.86512   Prob > F        =    0.0000
    Residual |  12874.1651    61,133  .210592725   R-squared       =    0.1577
-------------+----------------------------------   Adj R-squared   =    0.1576
       Total |  15283.8953    61,135  .250002377   Root MSE        =    .4589

------------------------------------------------------------------------------
   high_wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       yeduc |   .0489753   .0004608   106.29   0.000     .0480722    .0498785
         nin |  -.0123811   .0016981    -7.29   0.000    -.0157093   -.0090529
       _cons |   .0600717   .0059863    10.03   0.000     .0483385    .071805
------------------------------------------------------------------------------
```

2) Probit model

```
. probit high_wage yeduc nin

Iteration 0:  Log likelihood = -42376.037
Iteration 1:  Log likelihood = -37223.171
Iteration 2:  Log likelihood = -37222.747
Iteration 3:  Log likelihood = -37222.747

Probit regression                                 Number of obs   =     61,136
                                                  LR chi2(2)      =   10306.58
                                                  Prob > chi2     =     0.0000
Log likelihood = -37222.747                       Pseudo R2       =     0.1216

------------------------------------------------------------------------------
   high_wage | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
       yeduc |   .1370616   .0014279    95.99   0.000     .1342629    .1398603
         nin |  -.0346085    .004821    -7.18   0.000    -.0440573   -.0251596
       _cons |  -1.234311   .017857   -69.12   0.000    -1.26931   -1.199312
------------------------------------------------------------------------------
```

Logit model

```
. logit high_wage yeduc nin

Iteration 0:  Log likelihood = -42376.037
Iteration 1:  Log likelihood = -37211.498
Iteration 2:  Log likelihood =  -37208.51
Iteration 3:  Log likelihood =  -37208.51

Logistic regression                               Number of obs   =     61,136
                                                  LR chi2(2)      =   10335.05
                                                  Prob > chi2     =     0.0000
Log likelihood = -37208.51                        Pseudo R2       =     0.1219

------------------------------------------------------------------------------
   high_wage | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
       yeduc |   .2259258   .0024712    91.42   0.000     .2210822    .2307693
         nin |  -.0591557   .0080126    -7.38   0.000    -.0748682   -.0434432
       _cons |  -2.034067   .0302691   -67.20   0.000    -2.093394   -1.974741
------------------------------------------------------------------------------
```

Table:

Dependent variable: high_wage =1 if wage≥192000, =0 if wage<19200			
regression model	LPM	Probit	Logit
yeduc	0.0489753	0.1370616	0.2259258
	(0.0004608)	(0.0014279)	(0.0024712)
nin	-0.0123811	-0.0346085	-0.0591557
	(0.0016981)	(0.004821)	(0.0080126)
constant	0.0600717	-1.234311	-2.034067
	(0.0059863)	(0.017857)	(0.0302691)

Coefficient of yeduc in the linear probability model:

It means for each additional year of education, holding other factors constant, the probability of having a high wage larger than 192000 would increase by about 4.9%

2. ① Using a linear probability model may estimate the probability value greater than one or less than zero

② The linear probability model must exhibit heteroskedasticity

③ The estimated partial effects are constant for all values of independent variables

3.

```
. probit high_wage yeduc nin

Iteration 0:  Log likelihood = -42376.037
Iteration 1:  Log likelihood = -37223.171
Iteration 2:  Log likelihood = -37222.747
Iteration 3:  Log likelihood = -37222.747

Probit regression                                 Number of obs   =     61,136
                                                  LR chi2(2)      =   10306.58
                                                  Prob > chi2     =     0.0000
Log likelihood = -37222.747                       Pseudo R2       =     0.1216

------------------------------------------------------------------------------
   high_wage | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
       yeduc |   .1370616   .0014279    95.99   0.000     .1342629    .1398603
         nin |  -.0346085    .004821    -7.18   0.000    -.0440573   -.0251596
       _cons |  -1.234311   .017857   -69.12   0.000    -1.26931   -1.199312
------------------------------------------------------------------------------
```

According to the chart,

$\hat{\beta}_1 = 0.1370616$

$\hat{\beta}_2 = -0.0346085$

$\hat{\beta}_0 = -1.234311$

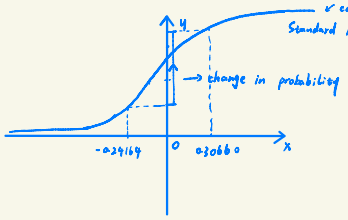
For the logit model $P(Y_i=1 \mid X_i, Z_i) = \Phi(\beta_0 + \beta_1 X_i + \beta_2 Z_i)$

$\therefore P(Y_i=1 \mid X=8, Z=3) = \Phi(-0.24164) = 0.404528$

$P(Y_i=1 \mid X=12, Z=3) = \Phi(0.30660) = 0.620427$

$\therefore$ The change in probability when $Z=3$ and X increases from 8 to 12 is $\Phi(0.30660) - \Phi(-0.24164) \approx 0.2159$

4.



↳ cdf of N(0,1)

Standard Normal Cumulative Distribution Function

Q3

1.

```
. regress wage nsgn_level

      Source |       SS           df       MS      Number of obs   =    61,136
-------------+----------------------------------   F(1, 61134)     =   4561.41
       Model |  1.8076e+14         1  1.8076e+14   Prob > F        =    0.0000
    Residual |  2.4227e+15    61,134  3.9629e+10   R-squared       =    0.0694
-------------+----------------------------------   Adj R-squared   =    0.0694
       Total |  2.6034e+15    61,135  4.2585e+10   Root MSE        =    2.0e+05

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
  high_level |   112764.3   1669.638     67.54   0.000     109491.8    116036.8
       _cons |   156631.9   1327.543    117.99   0.000     154029.9    159233.9
------------------------------------------------------------------------------
```

The coefficient of high-level is estimated to be 112764.3

Interpretion: It indicates that individuals with high education (≥ 9 years) level would earn 112764.3 more than individuals with no high level education on average, holding other factors constant.

2. i) $wage = \beta_0 + \beta_1 \text{high_level} + \beta_2 \text{urban} + \beta_3 \text{urban} \cdot \text{high_level} + \mu$

```
. regress wage high_level urban interaction

      Source |       SS           df       MS      Number of obs   =    61,136
-------------+----------------------------------   F(3, 61132)     =   1727.11
       Model |  2.0342e+14         3  6.7805e+13   Prob > F        =    0.0000
    Residual |  2.4000e+15    61,132  3.9259e+10   R-squared       =    0.0781
-------------+----------------------------------   Adj R-squared   =    0.0781
       Total |  2.6034e+15    61,135  4.2585e+10   Root MSE        =    2.0e+05

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
  high_level |   99802.76   1914.016     52.14   0.000     96051.29    103554.2
       urban |   25014.22   3553.394      7.04   0.000     18049.56    31978.88
 interaction |   23411.8    4131.893      5.67   0.000     15313.28    31510.32
       _cons |   152485.9   1446.661    105.41   0.000     149650.4    155321.3
------------------------------------------------------------------------------
```

$\therefore \hat{\beta}_0 = 152485.9$

$\hat{\beta}_1 = 99802.76$ → This is the extra monthly wage on average for those having high education and born in rural place.

$\hat{\beta}_2 = 25014.22$ → This is the difference of monthly wage on average between urbon born people with low education level and rural born people with low level education

$\hat{\beta}_3 = 23411.8$ → This the extra monthly wage for urban born people with high education level compared with rural born people with low education level.

2) Make the null hypothesis : $H_0 : \beta_3 = 0$: Birthplace can't affect the slope of high-level on wage

The alternative hypothesis : $H_1 : \beta_3 \neq 0$: Birthplace can affect the slope of high-level on wage

If we choose the significance level as 5%

∵ p-value for the interaction term high-level & urban is $0.000 < 5\%$

∴ We can reject H_0

∴ Birthplace can affect the slope of high-level on wage